



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

SECP2523 – DATABASE

SEM 1 - 2024/2024

NORMALIZATION

KADA COOPERATIVE SYSTEM

Group Name: KodeLab

Members:

1. LUBNA AL HAANI BINTI RADZUAN (A23CS0107) (LEADER)
2. NUR FIRZANA BINTI BADRUS HISHAM (A23CS0156)
3. NABIL AFLAH BOO BINTI MOHD YOSUF BOO YONG CHONG (A23CS0252)
4. NURUL ADRIANA BINTI KAMAL JEFRI (A23CS0258)
5. PRAVINRAJ A/L SIVABATHI (A23CS0171)

Submission Date :

1. 1NF:

Member(memberNo, icNo, password, role, applicationID)

PK: memberNo

FK: memberNo REFERENCES Membership(staffNo)

FK: icNo REFERENCES Membership(icNo)

FK: applicationID REFERENCES Membership_Notif(applicationID)

2NF:

- **No Partial Dependencies**

3NF:

- **No Transitive Dependencies**

2. 1NF:

StockRecord(memberNo, recordDate, shareCapital, feeCapital, fixedSavings, memberSavings, memberFund, adminNo)

PK: memberNo, recordDate

FK: memberNo REFERENCES Member(memberNo)

FK: adminNo REFERENCES Member(memberNo)

2NF:

- **No Partial Dependencies**

3NF:

- **No Transitive Dependencies**

3. 1NF:

AccountKoperasiRecord(rekodID, jualanBersih, kosJualan, jumlahPendapatan, jumlahPembelajaan, tarikhRekod, adminNo)

PK: rekodID

FK: adminNo REFERENCES Member_Stock(memberNo)

2NF:

- **No Partial Dependencies**

3NF:

- **No Transitive Dependencies**

4. 1NF:

Membership_Notif(applicationID, email, name, icNo, marriageStatus, homeAddress, homePostcode, homeCity, homeState, gender, religion, race, monthlySalary, staffNo, pfNo, positionGrade, workAddress, workPostcode, workCity, workState, faxNo, phoneNo, homePhoneNo, entranceFee, shareCapital, feeCapital, memberDeposit, charityFund, fixedSavings, membershipStatus, applicationDate, applicationStatus, approvalDate, comment, message, notificationDate, notificationStatus, adminNo)

PK: applicationID

FK: adminNo REFERENCES Member(memberNo)

2NF:

- **No Partial Dependencies**

3NF:

- **No partial dependencies**

Membership_Notif(applicationID, email, name, icNo, marriageStatus, homeAddress, homePostcode, homeCity, homeState, gender, religion, race, monthlySalary, staffNo, pfNo, positionGrade, workAddress, workPostcode, workCity, workState, faxNo, phoneNo, homePhoneNo, entranceFee, shareCapital, feeCapital, memberDeposit, charityFund, fixedSavings, membershipStatus, applicationDate, applicationStatus, message, notificationDate, notificationStatus, approvalDate, comment, adminNo)

PK: applicationID

FK: adminNo REFERENCES Member(memberNo)

5. 1NF:

FamilyInfo(familyMemberIc, familyMemberName, familyMemberBirthCertNo, familyMemberRelationship, applicationID)

PK: familyMemberIc

FK: applicationID REFERENCES Membership_Notif(applicationID)

2NF:

- **No Partial Dependencies**

3NF:

- **No Transitive Dependencies**

6. 1NF:

Loan_Notif(loanApplicationID, memberNo, loanType, loanAmount, loanPeriod, monthlyInstallment, dob, age, bankName, bankAccNo, verificationFile, loanApplicationStatus, message, notificationDate, notificationStatus, approvalDate, comment, adminNo)

PK: loanApplicationID

FK: memberNo REFERENCES Member(memberNo)

FK: adminNo REFERENCES Member(memberNo)

2NF:

- **No Partial Dependencies**

3NF:

- bankAccNo can determine bankName

BankDetail(bankAccNo, BankName)

PK: bankAccNo

Loan_Notif(loanApplicationID, memberNo, loanType, loanAmount, loanPeriod, monthlyInstallment, dob, age, bankAccNo, verificationFile, loanApplicationStatus, message, notificationDate, notificationStatus, approvalDate, comment, adminNo)

PK: loanApplicationID

FK: memberNo REFERENCES Member(memberNo)

FK: adminNo REFERENCES Member(memberNo)

FK: bankAccNo REFERENCES BankDetail(bankAccNo)

7. 1NF:

Guarantor(guarantorICNo, guarantorName, loanApplicationID)

PK: guarantorICNo

FK: loanApplicationID REFERENCES Loan_Notif(loanApplicationID)

2NF:

- **No Partial Dependencies**

3NF:

- **No Transitive Dependencies**

LAMA

1. **User_SAB_stock**(memberNo,email,password,role,shareCapital,feeCapital,fixedCapital,fixedSavings,memberSavings,memberFund)

➤ 1NF:

- User_SAB_stock(memberNo,email,password,role,shareCapital,feeCapital,fixedCapital,fixedSavings,memberSavings,memberFund)
- PK: memberNo
- FK: memberNo REFERENCES Membership(staffNo)
- FK: email REFERENCES Membership_Notif(email)

➤ 2NF:

- no partial dependency
- User_SAB_stock(memberNo,email,password,role,shareCapital,feeCapital,fixedCapital,fixedSavings,memberSavings,memberFund)
- PK: memberNo
- FK: memberNo REFERENCES Membership(staffNo)
- FK: email REFERENCES Membership_Notif(email)

➤ 3NF:

- transitive dependency
- User_stock(memberNo,email,shareCapital,feeCapital,fixedCapital,fixedSavings,memberSavings,memberFund)
- PK: memberNo
- FK: memberNo REFERENCES Membership(staffNo)
- FK: email REFERENCES User_SAB(email)
- User_SAB(email,password,role)
- PK: email
- FK: email REFERENCES Membership_Notif(email) (WHERE applicationStatus == Approved)

2. **Membership_Notif**(applicationID, email, name, icNo, marriageStatus, homeAddress, homePostcode, homeState, homeCity, gender, religion, race, monthlySalary, staffNo, pfNo, position, positionGrade, workAddress, workPostcode, workState, workCity, faxNo, phoneNo, homePhoneNo, entranceFee, shareCapital, feeCapital, memberDeposit, charityFund, fixedSavings, membershipStatus, applicationDate, applicationStatus, message, notificationDate, isAutomated, adminNo)

PK: applicationID

FK: adminNo REFERENCES User_SAB_Stock (adminNo)

- 1NF (already in its form)
- 2NF (already in its form)
- 3NF (look for transitive dependencies)
 - 1) homePostcode, homePhoneNo, homeState, and homeCity depend on homeAddress
 - 2) workPostcode, faxNo, workState, and workCity depend on workAddress

a] **Membership**(applicationID, email, name, icNo, marriageStatus, gender, religion, race, monthlySalary, staffNo, pfNo, position, positionGrade, entranceFee, shareCapital, feeCapital, memberDeposit, charityFund, fixedSavings, membershipStatus, applicationDate, applicationStatus, adminNo) **pfNo or staffNo**

PK: applicationID

FK: adminNo REFERENCES User_SAB_Stock (memberNo)

b] **HomeInfo Table**

HomeInfo(applicationID, homeAddress, homePostcode, homeState, homePhoneNo, homeCity)

PK: applicationID

FK: applicationID REFERENCES Membership(applicationID)

c] WorkInfo Table

WorkInfo(applicationID, workAddress, workPostcode, faxNo, workState, workCity, homePhoneNo)

PK: applicationID

FK: applicationID REFERENCES Membership(applicationID)

d] Notifications Table

Notifications(applicationID, message, notificationDate, isAutomated)

PK: applicationID

FK: applicationID REFERENCES Membership(applicationID)

3. **FamilyInfo**(familyMemberIC, applicationID, familyMemberName, familyMemberCertNo, familyMemberBirthRelationship)
PK: familyMemberIC, applicationID
FK: applicationID REFERENCES Membership_Notif (applicationID)

- 1NF (already in its form)
- 2NF (look for partial dependencies)
 - 1) familyMemberName, familyMemberCertNo, and familyMemberBirthRelationship depend only on familyMemberIC, not the full composite key.

a] *FamilyMember Table*

FamilyMember(familyMemberIC, familyMemberName, familyMemberCertNo, familyMemberBirthRelationship)
PK: familyMemberIC

b] *FamilyApplication Table*

FamilyApplication(familyMemberIC, applicationID)
PK: familyMemberIC, applicationID
FK: familyMemberIC REFERENCES FamilyMember(familyMemberIC)
FK: applicationID REFERENCES Membership_Notif(applicationID)

- 3NF (look for transitive dependency; none)

4. Review_Membership

1NF:

Review_Membership(bodNo, applicationID, reviewDate, comment)

PK: applicationID

FK: bodNo REFERENCES User_SAB_Stock(memberNo)

FK: applicationID REFERENCES Membership_Notif(applicationID)

2NF:

Review_Membership(bodNo, applicationID, reviewDate, comment)

PK: applicationID

FK: bodNo REFERENCES User_SAB_Stock(memberNo)

FK: applicationID REFERENCES Membership_Notif(applicationID)

3NF:

no transitive dependencies

Review_Membership(bodNo, applicationID, reviewDate, comment)

PK: applicationID

FK: bodNo REFERENCES User_SAB_Stock(memberNo)

FK: applicationID REFERENCES Membership_Notif(applicationID)

5. Loan_Notif

1NF

Loan_Notif(loanApplicationID, memberNo, loanType, loanAmount, loanPeriod, monthlyInstallment, dob, age, bankName, bankAccNo, loanApplicationStatus, message, notificationDate, isAutomated, adminNo)

PK: loanApplicationID

FK: memberNo REFERENCES User_SAB_Stock(memberNo)

FK : adminNo REFERENCES User_SAB_Stock(adminNo)

2NF (as all non-key attributes are fully depends on PK, no partial dependencies)

3NF

Member_Info(memberNo, dob, age, bankName, bankAccNo)

PK: memberNo

Loan_Notif(loanApplicationID, memberNo, loanType, loanAmount, loanPeriod, monthlyInstallment, loanApplicationStatus, message, notificationDate, isAutomated, adminNo)

PK: loanApplicationID, memberNo

FK: memberNo REFERENCES Member_Info(memberNo)

FK: adminNo REFERENCES User_SAB_Stock(memberNo)

PK: bodNo, applicationID

FK: bodNo REFERENCES Reviewer_Comments(bodNo)

FK: applicationID REFERENCES Review_Details(applicationID)

6. **Guarantor**(guarantorICNo, guarantorName, guarantorPFNo, guarantorMemberNo, guarantorPhoneNo, loanApplicationID)

➤ INF:

- Guarantor(guarantorICNo, guarantorName, guarantorPFNo, guarantorMemberNo, guarantorPhoneNo, loanApplicationID)
- PK: guarantorICNo, loanApplicationID
- FK: loanApplicationID REFERENCES Loan_Notif(loanApplicationID)

➤ 2NF:

guarantorICNo can determine guarantorName, guarantorPFNo, guarantorMemberNo, guarantorPhoneNo

- GuarantorInfo(guarantorICNo, guarantorName, guarantorPFNo, guarantorMemberNo, guarantorPhoneNo)
- PK: guarantorICNo
- Guarantor(guarantorICNo, loanApplicationID)
- PK: guarantorICNo, loanApplicationID
- FK: guarantorICNo REFERENCES GuarantorInfo(guarantorICNo)
- FK: loanApplicationID REFERENCES Loan_Notif(loanApplicationID)

➤ 3NF:

- GuarantorInfo(guarantorICNo, guarantorName, guarantorPFNo, guarantorMemberNo, guarantorPhoneNo)
- PK: guarantorICNo
- Guarantor(guarantorICNo, loanApplicationID)
- PK: guarantorICNo, loanApplicationID
- FK: guarantorICNo REFERENCES GuarantorInfo(guarantorICNo)
- FK: loanApplicationID REFERENCES Loan_Notif(loanApplicationID)

7. Review_Loan

1NF

Review_Loan(bodNo, loanApplicationID, reviewDate, comment)

PK: applicationID

FK: bodNo REFERENCES User_SAB_Stock(memberNo)

FK: loanApplicationID REFERENCES Loan_Notif(loanApplicationID)

2NF

Review_Loan(loanApplicationID, reviewDate, comment)

PK : loanApplicationID

FK : bodNo REFERENCES User_SAB_Stock(memberNo)

FK : loanApplicationID REFERENCES Loan_Notif(loanApplicationID)

3NF

Review_Loan(loanApplicationID, reviewDate, comment)

PK : loanApplicationID

FK : bodNo REFERENCES User_SAB_Stock(memberNo)

FK : loanApplicationID REFERENCES Loan_Notif(loanApplicationID)